

Session 3: Data Types and VariablesTopics:Data types (int, float, char, double).Variable naming rules.Constants, literals.Example: Store and print student's name, age, marks.

Tasks

1.

Declare variables for a Flipkart product: productName (as a string), price (float), and rating (double). Assign sample values and print each variable with its data type.

```
#include <stdio.h>

int main()
{
    char productName[] = "Samsung Mobile";

    float price = 24999.50;

    double rating = 4.5;

    printf("Product Name: %s\n", productName);

    printf("Data Type: char array\n\n");

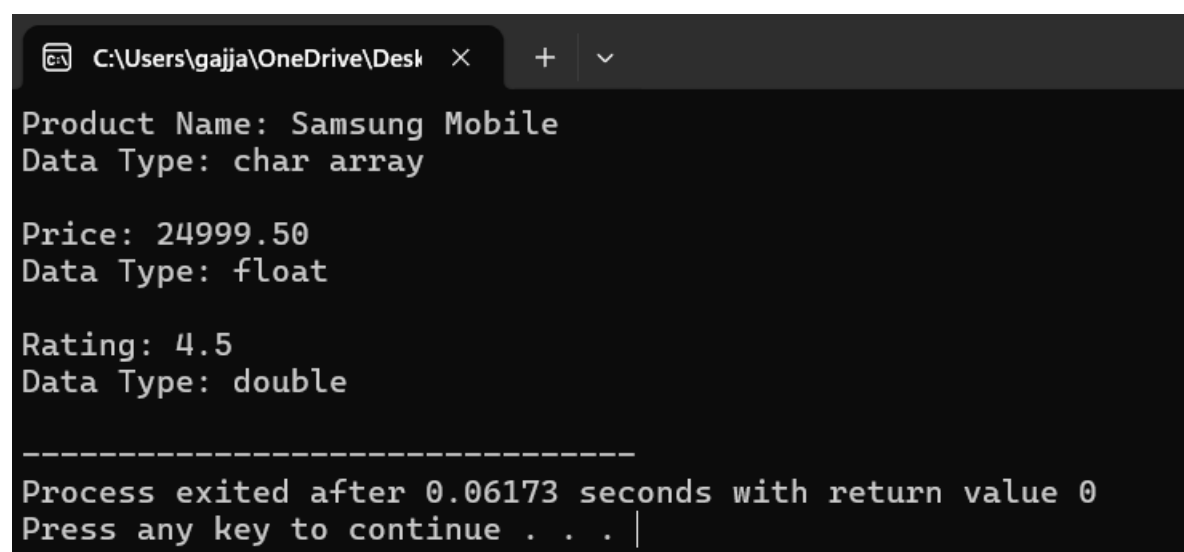
    printf("Price: %.2f\n", price);

    printf("Data Type: float\n\n");

    printf("Rating: %.1lf\n", rating);

    printf("Data Type: double\n");

    return 0;
}
```



```
C:\Users\gajja\OneDrive\Desktop >
Product Name: Samsung Mobile
Data Type: char array

Price: 24999.50
Data Type: float

Rating: 4.5
Data Type: double

-----
Process exited after 0.06173 seconds with return value 0
Press any key to continue . . . |
```

2.

Create a constant variable to store the GST rate (for example, 18%) and use it to calculate the final price of a Zomato order with a given base price.

Constraint: The GST rate must not be changeable after its initial assignment.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    const float GST_RATE = 18.0;
```

```
    float basePrice = 500.00;
```

```
    float gstAmount;
```

```
    float finalPrice;
```

```
    gstAmount = basePrice * GST_RATE / 100;
```

```
    finalPrice = basePrice + gstAmount;
```

```
    printf("Base Price: %.2f\n", basePrice);
```

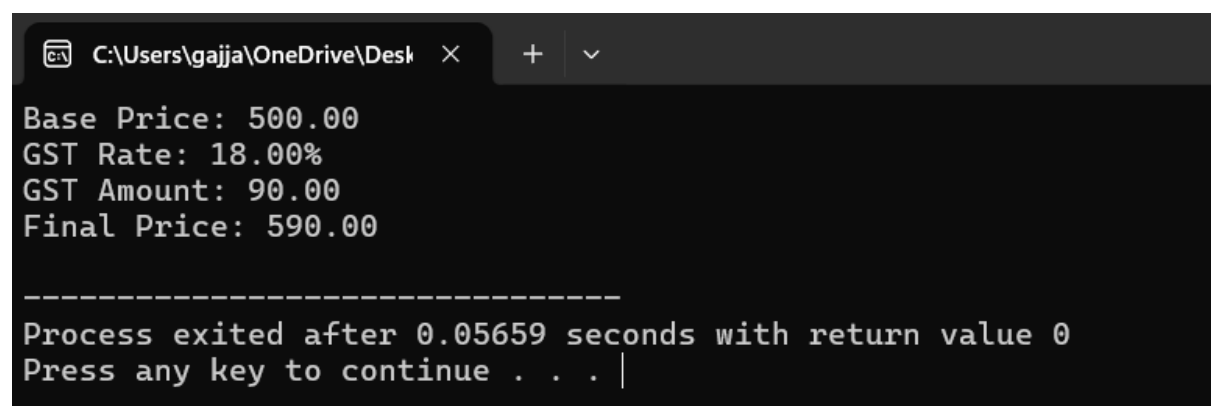
```
    printf("GST Rate: %.2f%%\n", GST_RATE);
```

```
    printf("GST Amount: %.2f\n", gstAmount);
```

```
    printf("Final Price: %.2f\n", finalPrice);
```

```
    return 0;
```

```
}
```



```
C:\Users\gajja\OneDrive\Desktop >
Base Price: 500.00
GST Rate: 18.00%
GST Amount: 90.00
Final Price: 590.00

-----
Process exited after 0.05659 seconds with return value 0
Press any key to continue . . . |
```

3.

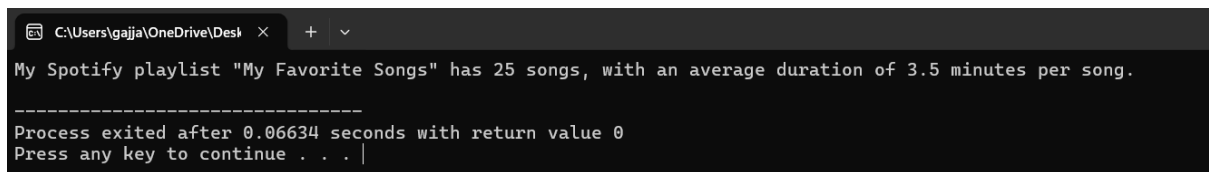
Write a program that stores your favorite Spotify playlist's name (string), total number of songs (int), and average song duration in minutes (float). Print all values in a single formatted sentence.

```
#include <stdio.h>
```

```
int main()
{
    char playlistName[] = "My Favorite Songs";
    int totalSongs = 25;
    float averageDuration = 3.5;

    printf("My Spotify playlist \"%s\" has %d songs, "
           "with an average duration of %.1f minutes per song.\n",
           playlistName, totalSongs, averageDuration);

    return 0;
}
```



```
C:\Users\gajja\OneDrive\Desk >
My Spotify playlist "My Favorite Songs" has 25 songs, with an average duration of 3.5 minutes per song.
-----
Process exited after 0.06634 seconds with return value 0
Press any key to continue . . .
```

4.

Identify and correct the invalid variable names from this list: 1stPlayer, player_age, \$score, total-marks, userName.

Hint: Apply standard variable naming rules for your programming language.

Variable	Valid?	Correct version
1stPlayer	 Invalid	firstPlayer
player_age	 Valid	player_age
\$score	 Invalid	score
total-marks	 Invalid	total_marks
userName	 Valid	userName